

Mark Scheme

Q1.

	Scheme	Marks	AO
(a)	Negative	B1 (1)	1.2
(b)(i)	Rainfall	B1	2.2b
(ii)	mm <u>or</u> Pressure hPa or Pascals or hectopascals or mb or millibars	B1ft (2)	1.1b
(c)	$H_0 : \rho = 0$ $H_1 : \rho \neq 0$ Critical value: $-0.361(0)$ $r < -0.3610$ so significant result and there is evidence of a correlation between Daily Total <u>Sunshine</u> and Daily Maximum Relative <u>Humidity</u>	B1 M1 A1 (3)	2.5 1.1b 2.2b
(d)	Humidity is high and there is evidence of correlation and $r < 0$ So expect amount of sunshine to be <u>lower</u> than the <u>average</u> for Heathrow(oe)	B1 (1)	2.2b
		(7 marks)	

Notes	
(a)	B1 for stating negative. "Negative skew" is B0 though
(b)(i)	B1 for mentioning "rainfall" (allow "rain" or "precipitation") or "pressure" (if more than 1 answer both must be correct) NB the other quantitative variable for Perth is: Daily Mean Wind Speed and scores B0 [Not allowed "wind speed" since $r = +0.15$ and in winter might expect wind to raise temp]
(ii)	B1ft for giving the correct units. If Daily Mean Wind Speed (kn) or knots "Wind speed" and "knots" would score B0B1 but any other variable scores B0B0
(c)	B1 for both hypotheses correct in terms of ρ M1 for the correct critical value compatible with their H_1 : allow $\pm 0.361(0)$ If the hypotheses are 1-tail then allow cv of ± 0.3061 e.g. Alternative hypothesis with $r < \pm 0.377$ implies a one-tail test or H_0 and H_1 in words saying " H_0 : there is no correlation, H_1 : there is correlation" is two-tail If there are no hypotheses (or they are nonsensical) assume 2-tail so M1 for $\pm 0.361(0)$ A1 for a correct conclusion in context based on comparing -0.377 with their cv. Condone incorrect inequality e.g. $-0.3610 < -0.377$ as long as they reject H_0 Do not accept contradictory statements such as "accept H_0 so there is evidence of ..." Can say "support for Stav's belief" (o.e.e.g. "claim") or "evidence of a correlation between <u>sunshine</u> and <u>humidity</u> " condone "negative correlation" or comments such as "if humidity is high amount of sunshine will be low"
(d)	B1 for stating <u>low</u> amount of sunshine (o. e.) and some reference to $r < 0$ or fog Check for the following 2 features: (i) low sunshine: allow ≤ 5 hrs (LDS mean for 2015 is 5.3, humidity 97% is 4.1, $\geq 97\%$ is 3.1) (ii) negative correlation may be described in words e.g. "high humidity gives low sunshine" <u>or fog</u> (LDS says $>95\%$ humidity is foggy) so less sunshine

Q2.

	Scheme	Marks	AO
(a)	Negative	B1 (1)	1.2
(b)	Marc's suggestion is <u>compatible</u> because it's <u>negative correlation</u>	B1 (1)	2.4
(c)	$(r =) -0.54458266...$ awrt <u>-0.545</u>	B1 (1)	1.1b
(d)	$H_0: \rho = 0$ $H_1: \rho < 0$ [5% 1-tail cv =] $(\pm) 0.4259$ (significant result / reject H_0) There is evidence of negative <u>correlation</u> between the <u>number of letters</u> in (or <u>length</u> of) a student's last <u>name</u> and their first <u>name</u>	B1 M1 A1 (3)	2.5 1.1a 2.2b
		(6 marks)	

	Notes
(a)	B1 for "negative" Allow "slight" or "weak" etc Allow a description e.g. "as x increases y decreases" or in context e.g. "people with longer last names tend to have shorter first names" A comment of "negative skew" is B0 Need to see distinct or separate responses for (a) and (b)
(b)	B1 for a comment that suggests data is compatible with the suggestion and a suitable reason such as "there is negative correlation" <u>or</u> a description in x and y or in context <u>or</u> the points lie close to a line with <u>negative gradient</u> <u>or</u> draw line $y = x$ and state that <u>more points below the line</u> so <u>supports (or is compatible with) his suggestion</u> A reason based on just a single point is B0 e.g. "11 letters in last name has only 5 in first name"
(c)	B1 for awrt -0.545
(d)	B1 for both hypotheses correct in terms of ρ M1 for a critical value compatible with their H_1: 1-tail: awrt ± 0.426 (condone ± 0.425) or 2-tail (B0 scored for H_1): awrt ± 0.497 If hypotheses are in words and can deduce whether one or two-tail then use their words. If no hypotheses or their H_1 is not clearly one or two tail assume one-tail A1 for compatible signs between cv and r and a correct conclusion in context mentioning <u>correlation</u> and <u>number of letters</u> or <u>length</u> and <u>name</u> (ft their value from (c)) Do NOT award this A mark if contradictory comments or working seen e.g. "accept H_0" or comparison of 0.426 with significance level of 0.05 etc The M1A1 can be scored independently of the hypotheses
NB	

Q3.

Question	Scheme	Marks	AOs
(a)	eg As the number of minutes <u>exercise</u> (m) increases the resting <u>heart rate</u> (h) decreases or the gradient of the curve is becoming flatter with increasing m : diminishing effect of each <u>additional minute of exercise</u>	B1	2.4
		(1)	
(b)	$H_0 : \rho = 0$ $H_1 : \rho < 0$	B1	2.5
	Critical value -0.3887 (Allow $\pm$)	M1	1.1b
	There is evidence that the product moment <u>correlation</u> is <u>less than 0</u> / <u>there is a negative correlation</u>	A1	2.2b
		(3)	
(c)	$\log_{10} h = -0.05 \log_{10} m + 1.92$ $h = am^k \rightarrow \log_{10} h = \log_{10} am^k$	M1	1.1b
	$\log_{10} h = -\log_{10} m^{0.05} + 1.92$ or $\log_{10} h = \log_{10} m^{-0.05} + 1.92$ or $h = 10^{1.92-0.05 \log_{10} m}$ oe	M1	2.1
	$\log_{10} hm^{0.05} = 1.92$ or $\log_{10} \left(\frac{h}{m^{-0.05}} \right) = 1.92$ or $h = 10^{1.92} \times 10^{-0.05 \log_{10} m}$ oe	M1	1.1b
	$hm^{0.05} = 10^{1.92}$ or $\frac{h}{m^{-0.05}} = 10^{1.92}$ or $h = 10^{1.92} \times 10^{\log_{10} m^{-0.05}}$	M1	1.1b
	$h = 10^{1.92} m^{-0.05}$ or $h = 83.17...m^{-0.05}$ or $a = \text{awrt } 83.17$ and $k = -0.05$	A1	1.1b
		(5)	

Notes:		(9 marks)
(a)	B1	eg Idea as one increases the other decreases (in context). Allow use of m and h eg As m increases h decreases. Do not allow negative correlation with no context or $\rho < 0$ Allow there is a negative correlation/association/relationship/exponential between minutes <u>exercise</u> (m) and resting <u>heart rate</u> (h) oe
(b)	B1	Both hypotheses correct in terms of ρ (allow p)
	M1	For the cv of -0.3887 or any cv such that $0.3 < cv < 0.5$
	A1	Independent of hypotheses. Correct conclusion that implies reject H_0 on basis of seeing -0.3887 or if they give 0.3887 we must see the comparison $0.3887 < 0.897$ and which mentions "pmcc/correlation/relationship" and less than 0/ negative or $\rho < 0$ A contradictory statement scores A0 eg Accept H_0 therefore negative correlation
(c)		In this part once M0 is scored no more marks can be scored. Condone no base
	M1	May be implied by 2nd M1 mark Method 1: Correct substitution for both x and y Method 2 : Taking the log of both sides
	M1	May be implied by 3rd M1 mark Method 1: Correct use of the power log rule or making h the subject Method 2 : Correct use of the addition/subtraction log rule
	M1	This line implies M1M1M1 Method 1: Correct use of the addition/subtraction log rule or eq ⁿ in the form $h = 10^{1.92} \times 10^{-0.05 \log m}$ Method 2: A second correct step for correct use of the power log rule
	M1	This line implies M1M1M1M1 Method 1: Correct removal of logs or $h = 10^{1.92} \times 10^{\log m^{-0.05}}$ Method 2: Log a (or a) and k correct
	A1	Allow $h = \text{awrt } 83.2m^{-0.05}$ NB award 5/5 for $a = \text{awrt } 83.2$ and $k = -0.05$ or $h = \text{awrt } 83.2...m^{-0.05}$ or $h = 10^{1.92} m^{-0.05}$

Q4.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$H_0 : \rho = 0$ $H_1 : \rho > 0$	B1	This mark is given for both hypotheses in terms of ρ found correctly
	For sample size 24 at the 5% level of significance, the critical value = 0.3438	M1	This mark is given for selecting a suitable critical value compatible with H_1
	$0.446 > 0.3438$, so reject H_0 There is evidence that the product moment correlation coefficient (pmcc) is greater than 0	A1	This mark is given for a correct conclusion stated
(b)	The value of the pmcc is close to 1 so there is a strong positive correlation	B1	This mark is given for a correct explanation about the strength of the correlation
(c)	$\log_{10} y = -1.82 + 0.89 \log_{10} x$	M1	This mark is given for a correct substitution of both c and m
	$y = 10^{-1.82 + 0.89 \log x}$	M1	This mark is given for dealing with logs to find an expression in terms of y
	$y = 10^{-1.82} \times 10^{0.89 \log x}$ $y = 10^{-1.82} \times 10^{(\log x)^{0.89}}$	M1	This mark is given for a method to find values for a and n
	$y = 0.015 \times x^{0.89}$	A1	This mark is given for find a correct value of $a = 0.015$
		A1	This mark is given for find a correct value of $n = 0.89$
(Total 9 marks)			

Q5.

Qu	Scheme	Marks	AO
(a)	$H_0 : \rho = 0$ $H_1 : \rho < 0$ Critical value: -0.6215 (Allow any cv in range $0.5 < cv < 0.75$) $r < -0.6215$ so significant result and there is evidence of a negative correlation between w and t	B1 M1 A1 (3)	2.5 1.1a 2.2b
(b)	e.g. As temperature increases people spend more time on the beach and less time shopping (o.e.)	B1 (1)	2.4
(c)	Since r is close to -1 , it is consistent with the suggestion	B1 (1)	2.4
(d)	t will be the explanatory variable since sales are likely to depend on the temperature	B1 (1)	2.4
(e)	Every degree rise in temperature leads to a drop in weekly earnings of £171	B1 (1)	3.4
		(7 marks)	
Notes			
(a)	B1 for both hypotheses in terms of ρ M1 for the critical value: sight of ± 0.6215 or any cv such that $0.5 < cv < 0.75$ A1 must reject H_0 on basis of comparing -0.915 with -0.6215 (if $-0.915 < -0.6215$ is seen then A0 but may use $ r $ o.e. which is fine) <u>and mention "negative", "correlation/relationship" and at least "w" and "t"</u>		
(b)	B1 for a suitable <u>reason to explain</u> negative correlation using the context given. e.g. "As temperature drops people are more likely to go shopping (than to the beach)" e.g. "As temperature increases people will be outside rather than in shops" A mere description in context of negative correlation is B0 SO e.g. "As temperature increases people don't want to go shopping/buy clothes" is B0 e.g. "Less clothes needed as temp increases" is B0		
(c)	B1 for a suitable reason e.g. "strong"/"significant"/"near perfect" "correlation", $ r $ close to 1 <u>and saying it is consistent with the suggestion. Allow "yes" followed by the reason.</u>		
(d)	B1 For identifying t <u>and</u> giving a suitable reason. Need idea that " w <u>depends on</u> t " or " w <u>responds to</u> t " or " t <u>affects</u> w " (o.e.) Allow t (temperature) <u>affects</u> the other variable etc Just saying " t is the independent variable" or " t <u>explains</u> change in w " is B0 N. B. Suggesting causation is B0 e.g. " t causes w to decrease"		
(e)	B1 for a description that conveys the idea of rate per degree Celsius. Must have 171, condone missing "£" sign.		

Q6.

Question	Scheme	Marks	AOs
(a)	e.g. It requires extrapolation so will be unreliable (o.e.)	B1	1.2
		(1)	
(b)	e.g. Linear association between w and t	B1	1.2
		(1)	
(c)	$H_0: \rho = 0$ $H_1: \rho > 0$	B1	2.5
	Critical value 0.5822	M1	1.1a
	Reject H_0		
	There is evidence that the product moment correlation coefficient is greater than 0	A1	2.2b
		(3)	
(d)	Higher $\bar{t}$ suggests overseas and not Perth...lower wind speed so perhaps not close to the sea so suggest Beijing	B1	2.4
		(1)	
(6 marks)			
Notes:			
(a)			
B1: for a correct statement (unreliable) with a suitable reason			
(b)			
B1: for a correct statement			
(c)			
B1: for both hypotheses in terms of ρ			
M1: for selecting a suitable 5% critical value compatible with their H_1			
A1: for a correct conclusion stated			
(d)			
B1: for suggesting Beijing with some supporting reason based on t or w			
Allow Jacksonville with a reason based just on higher $\bar{t}$			